



THE SURVEYING AND MAPPING ACT, 2014



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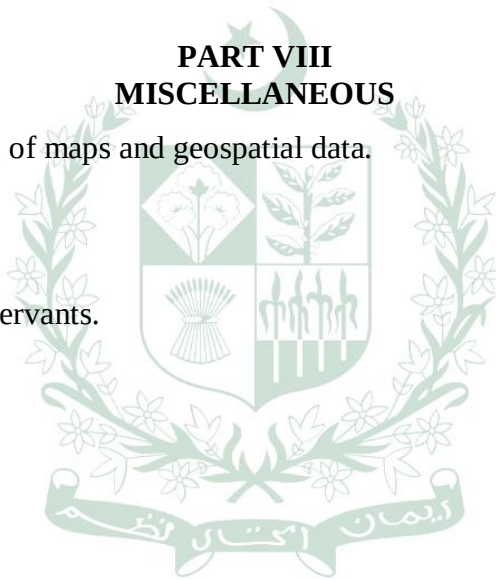
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THE SURVEYING AND MAPPING ACT, 2014

(ACT No. I OF 2014)

[12th May, 2014]

An Act to provide for constitution and regulation of survey of Pakistan

WHEREAS in the view of prevailing circumstances it is expedient to constitute Survey of Pakistan to provide for the regulation of surveying and mapping activities and for the matters connected therewith or ancillary thereto;

AND WHEREAS it is necessary to provide for speedy disposal of such cases and execution in matters connected therein or ancillary thereto;

It is hereby enacted as follows:—

PART I

PRELIMINARY

1. Short title, extent and commencement.—(1) This Act may called the Surveying and Mapping Act, 2014.

(2) It extends to the whole of Pakistan.

(3) It shall come into force at once.

2. Definitions.—In this Act, unless there is anything repugnant in the subject or context,—

- (i) "aerial photography" means the act of taking photographs of any portion of the earth with the help of cameras installed in an aircraft or any sort of air borne vehicle;
- (ii) "datum" means an arbitrarily selected reference point, plane or surface to which all measurements are referred including geodetic, vertical and gravimetric datum;
- (iii) "deposit works" mean extra departmental jobs carried out on payment basis;
- (iv) "digital maps" mean geographic maps produced through satellite images, GPS, total station data or digitized using mapping software;
- (v) "engineering maps" mean the large-scale topographical maps on which engineering works, of development projects are planned and designed, generally their scale is equivalent or larger than 1: 10,000;

- (vi) "gravity datum" means the point to which all gravity measurements are referred;
- (vii) "Government department" means a department controlled by Federal or a Provincial Government and includes any autonomous body, authority or organization established by or under the Federal or a Provincial law or which is owned or controlled or administered by the Federal or a Provincial Government or in which the Federal or a Provincial Government have controlling share or interest and also includes any local Government;
- (viii) "geodetic datum" means the origin or a point to which all horizontal measurements are referred;
- (ix) "geographical maps" mean maps that show general configuration of physical features of a certain area on a reference plane;
- (x) "geospatial data" means information collected in hard copy format or in digital format or any other format for the purposes of preparation of geographical or topographical data and its attributes, its processing and analysis;
- (xi) "imagery" means remotely sensed image of any resolution depicting any part of the earth and acquired or obtained using optical or, as the case may be, non-optical sensor;
- (xii) "local plane coordinate system" means a rectangular coordinate system generally used for large-scale mapping;
- (xiii) "mapping" means the art of map-making using specific datum, projection and scale and includes digital mapping;
- (xiv) "national co-ordinate system" means a plane coordinate system as notified by the ¹[concerned Division];
- (xv) "National Naming Authority" means the National Naming Authority established under section 19;
- (xvi) "National Spatial Data Infrastructure" means a system of geospatial databases handling facilities capable of interacting with each other for effective use of geospatial data and include set of standards, specifications and policies defining a framework for collecting, sharing, disseminating, processing and carrying out analysis on geospatial data in the most cost effective manner with the key features to allow unrestricted yet controlled sharing of data for decision support system at strategic, tactical and operational levels;
- (xvii) "prescribed" means prescribed by rules made under this Act;
- (xviii) "photogrammetry" means the art of taking measurements and mapmaking by using mono or, as the case may be, stereo image of the photographs taken by an aircraft or air borne vehicle or a satellite;

¹ Substituted by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.2.

- (xix) "public survey" means a survey in respect of which details are included in the register of public surveys;
- (xx) "registered surveyor" means a person who is registered with the Survey of Pakistan as a land surveyor under this Act and competent to carry out authorized type of survey work, processing and analysis;
- (xxi) "remote sensing" means the method and art by which characteristics of the objects of interest can be identified, classified, measured or analyzed without physical contact;
- (xxii) "registered organization" means any organization or authority or agency of the Government, a private firm or organization duly registered with the Survey of Pakistan under this Act and competent to carry out authorized type of survey work, processing and analysis;
- (xxiii) "survey" means the act of taking topographic surveying measurements and collection of spatial and non-spatial data for preparing geospatial database, its processing and analysis;
- (xxiv) "satellite" means an artificial body revolving around the earth at some distance;
- (xxv) "Survey of Pakistan" means the department of Survey of Pakistan, which is the national surveying and mapping organization of Pakistan;
- (xxvi) "Surveyor General of Pakistan" means head of the department of Survey of Pakistan;
- (xxvii) "survey mark" means benchmark pillar, iron bolt fixed in rock, a mark engraved on any building, a mark engraved on an in situ rock or a stone embedded in ground, a triangulation point protected by a cairn, a gravity pillar, a leveling benchmark, a fixed mark, hill station, monument or any other station or mark created by the Survey of Pakistan for the purpose of surveying and mapping;
- (xxviii) "sub-standard work" means any survey work which does not conform to the approved specifications and standards as may be prescribed by the Survey of Pakistan;
- (xxix) "topographical data" means data, which shows detailed information about shape, size and relationship of all features and abstract information of the earth, including but not limited to natural and manmade objects or features, geographical names, administrative boundaries and information like elevation, hydrology, vegetation, population and others; and

(xxx) "vertical datum" means the surface to which all vertical measurements are referred and mean sea level is taken as vertical datum of a country.

PART II

CONSTITUTION OF SURVEY OF PAKISTAN

3. Constitution of Survey of Pakistan.—(1) Notwithstanding anything contained in any other law for the time being in force, the Survey of Pakistan shall be deemed to have been established under this Act.

(2) For efficient administration of Survey of Pakistan, there shall be a Surveyor ¹[General] of Pakistan who shall be appointed by the ¹[Prime Minister] and there shall be other officers and staff as may be appointed ¹[****] in the prescribed manner.

¹[(3) Survey of Pakistan shall be technical department for all activities involving surveying, mapping, geographic information system or remote sensing and geospatial information technologies in the country.]

4. Offices of Survey of Pakistan.—Survey of Pakistan shall have its head office at Rawalpindi or Islamabad and shall have offices at Federal, Provincial or district level and such other offices as the Survey of Pakistan may, by notification in the Official Gazette, establish.

5. Administration of Survey of Pakistan.—Under this Act,—

- (a) superintendence of Survey of Pakistan shall vest in the Federal Government; and
- (b) administration of Survey of Pakistan shall vest in, and be exercised by, the Surveyor General of Pakistan in accordance with the provisions of this Act and the rules made thereunder.

6. Functions of Survey of Pakistan.—For carrying out the purposes of this Act, Survey of Pakistan shall—

- (a) be the sole national surveying and mapping organization of Pakistan;
- (b) determine and update geodetic datum and projection system of Pakistan using latest geodetic techniques and technologies;
- (c) determine and update vertical datum using data captured by Survey of Pakistan and obtained from Government departments, registered organizations or any other source having such installations along the coastal line of Pakistan;
- (d) delineate and demarcate international borders and relocate boundary pillars;

- (e) provide geodetic control, heights and geographical positions all over Pakistan;
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¹Subs., omitted and added by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s. 3.

- (f) advise ¹[Prime Minister] on the practices to be followed in the production of geospatial data as well as surveying and mapping practices and as the case may be in the collection and dissemination of this kind of information needed for geographical information systems (GIS) applications;
- (g) prepare geospatial data, remote sensing and geographical information system applications of the entire country on various scales as advised by the ¹[Prime Minister] and provide above services as a deposit work for a specific area with pre-defined specifications to the Provincial Governments, district and local governments and other clients from public as well as private sector;
- (h) carry out topographic survey, its updation and printing of topographic maps of national map series;
- (i) compile derived maps on various scales as required by the ¹[concerned Division];
- (j) generate maps on any scale through aerial photography and remote sensing using analytical and digital methods;
- (k) be responsible to prepare and print guide maps, provincial maps, district maps, road maps, tehsil maps and miscellaneous geographical maps for the whole of Pakistan;
- (l) establish and maintain National Spatial Data Infrastructure (NSDI) of Pakistan;
- (m) be responsible to survey and prepare cantonment areas maps;
- (n) train departmental employees and potential candidates from private sector in various disciplines of surveying, geographical and mapping in the Survey Training Institute at various levels;
- (o) control and coordinate surveying and mapping activities done by registered organizations and individuals and their registration process;
- (p) carry out magnetic survey of entire country in a cyclic order as prescribed by Survey of Pakistan for the purpose of magnetic declination;
- (q) have such other functions as are conferred or imposed on it by or under this Act or any other law; and

- (r) perform such other functions as may be assigned by the ¹[concerned Division].

¹ Substituted by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.4.

7. Research and development activities.—(l) Survey of Pakistan shall encourage and collaborate research activities in the field of surveying and mapping, remote sensing, GIS and other fields related to the discipline of surveying and mapping, cartography, instrument development and database with national and international organizations.

(2) Survey of Pakistan shall endeavor to upgrade scientific and technology level in this field and encourage and support any registered organizations and individuals that produce outstanding work in any field of surveying and mapping and related scientific or technological research.

8. Surveys and demarcation, etc.—(l) Survey of Pakistan shall be responsible for the delineation and demarcation of international land boundaries of Pakistan, jointly with the national mapping organizations or other organizations as authorized by the neighboring countries, in accordance with mutually agreed protocol and the plans drawn in the light of such protocol before taking up actual demarcation work.

(2) Survey of Pakistan shall carry out surveying and mapping of the provinces, districts and other administrative boundaries of Pakistan on receipt of requests there for after the alignment thereof has been agreed upon.

(3) Survey of Pakistan may also take up demarcation of land or state boundaries as a deposit work on mutual agreement of the parties consisting of Federal Government, Provincial Governments, district Governments, semi-government departments, autonomous bodies, private parties or individuals or as ordered by a court of law.

(4) Fifty *per cent* of the money received against the deposit shall be utilized for capacity building of the Survey of Pakistan in accordance with the prescribed procedure.

9. Authorization to enter into any land.—(1) Any person authorized by Survey of Pakistan for the purpose of this section may, in accordance with the applicable law,—

- (a) enter into any land and public building, office or structure for the purpose of carrying out assigned task of collection of spatial or nonspatial data or information subject to applicable laws;
- (b) take measurements in or from any such land from survey mark;
- (c) search for and, if necessary, uncover any existing survey marks in or on any such land; and
- (d) insert new survey marks in or on any such land for the purpose of carrying out any survey under this Act or for the purpose of maintaining or repairing any permanent survey marks.

(2) This section does not entitle any person so authorized to enter any building or structure on land, other than a building or structure referred to in clause (a) of sub-section (1).

10. Datum, projection and standards.—(1) Survey of Pakistan shall establish and adopt geodetic, vertical and gravimetric datum at national level. This datum shall be adopted in all national activities of surveying and mapping, including those carried out by the armed forces of Pakistan and the Geological Survey of Pakistan (GSP).

(2) Survey of Pakistan shall define a unified projection system and system of geodetic coordinates and plane co-ordinates and shall define classification, order and accuracies of these co-ordinate systems as may be prescribed.

(3) For the purpose of this section, Survey of Pakistan shall establish independent local plane co-ordinate systems to meet the needs of construction work, urban planning and scientific research activities, which systems may be connected with the national co-ordinate system and shall be kept and maintained as classified information.

11. Registration.—(1) No public or private organization, private firm or individual, national or international, shall ¹[use, insert, print or publish maps of Pakistan or atlases,] undertake any geospatial data collection, production or analysis work and surveying and mapping activities unless they are registered with Survey of Pakistan for such purpose as may be prescribed. The qualifications of staff and suitability for such work shall also be certified by Survey of Pakistan in the prescribed manner.

(2) Before registration, Survey of Pakistan shall obtain clearance of such firms, organizations and individuals in accordance with such procedure as may be prescribed.

(3) Sub-sections (1) and (2) shall not apply to operations of geospatial data production, analysis, surveying and mapping carried out by—

- (a) any student of surveying, studying in a public educational institution or a university; and
- (b) any person who produces geospatial data under immediate supervision of a registered organization, firm or individual.

12. Register.—(1) Survey of Pakistan shall maintain a register for the purpose of geospatial data of public surveys which shall contain details about—

- (a) surveys carried out by Survey of Pakistan; and
- (b) surveys for which details have been forwarded to Survey of Pakistan by any registered organization or individual.

(2) Survey of Pakistan may ensure that public survey conforms with the prescribed standards.

13. Joint survey and mapping and geospatial data production work with foreign companies.—No work related to surveying and mapping, collection and production

¹ Inserted by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.5

of geospatial data within Pakistan shall be undertaken by any individual, private firm or Government organization in collaboration with any foreign company or firm or non-governmental organization unless prior written approval is obtained from the ¹[concerned Division], observing the requisite channels.

14. Correction of erroneous data.—(1) Where it comes to knowledge of Survey of Pakistan that erroneous data is being or was produced by a registered individual, firm or organization, it—

- (a) may nominate an authorized official to inform by notice in writing requiring such data producer to correct the same within the time specified in the notice; and
- (b) shall have the authority to engage another registered surveyor to make the correction if data producer does not comply with the notice. The defaulting data producer shall be informed to this effect through a written notice.

(2) The data specified in sub-section (1), if not corrected within the time specified therein, shall immediately be forfeited by Survey of Pakistan and shall not be used for any purpose whatsoever.

(3) Any costs or expenses incurred on work mentioned in sub-section (1) shall, as a debt due to the Government, be recoverable through a court of competent jurisdiction from the registered individual, firm or organization surveyor.

PART III

MANAGEMENT OF GEOSPATIAL DATA

15. Management of geospatial data.—(1) Survey of Pakistan shall establish and maintain National Spatial Data Infrastructure (NSDI) with support of key stakeholders to ensure consistent mechanism of maintenance, dissemination and sharing geospatial data among all users by reducing duplication in collection and maintenance of aforesaid data and to enhance and improve objective decision making.

(2) NSDI established and maintained under sub-section (1) shall, for development of a system of geospatial databases and data handling facilities, be capable of interacting amongst all stakeholders and ensure putting in place design, implementation and maintenance mechanisms for facilitating, sharing, access and responsible for effective use of geospatial data at an affordable cost.

(3) All basic data or information comprising geodetic data, gravity data, magnetic data, topographical or geographical data, aerial photographs and all relevant records, both analogue and digital, prepared and possessed by Survey of Pakistan shall be maintained in the

form of a national geospatial database. Any registered organization, developing geospatial data, shall, in accordance with the procedure prescribed, forward a copy of the data to Survey

¹ Substituted by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.6.

of Pakistan free of cost for inclusion in the national geospatial database, except the classified data collected by army survey group engineers, Geological Survey of Pakistan or strategic organizations.

(4) Survey of Pakistan shall regularly update geospatial data in the national spatial data infrastructure and supply the same to various users on demand as may be prescribed.

PART IV

SUPPLY AND SECURITY OF GEOSPATIAL DATA

16. Supply and use of geospatial data.—(1) All geospatial data, aerial photographs and satellite imageries may be supplied in the manner and on payment to any individual or organization, as may be prescribed. Classified data shall be provided in the manner prescribed and the same shall be returned to Survey of Pakistan after the specified period.

(2) No classified maps and data shall be allowed to be used by foreign consultants, firms or organizations without prior written approval of the ¹[concerned Division].

(3) Without prior written approval of the Surveyor General of Pakistan, no individual, company, firm, private or Government agency shall copy, digitize or print a map or aerial photograph or a part thereof prepared by Survey of Pakistan except digitization by army survey group engineers for operational requirement of armed forces of Pakistan.

(4) Any map which is top secret, secret or restricted shall not be exported out of Pakistan in any format or by any means.

(5) Except as provided under this Act, any act of copying, digitizing or printing any geographic map or photograph or satellite imagery shall be prohibited.

¹[(6) No one shall print, display, disseminate, use or circulate incorrect and unofficial version of map of Pakistan or any part of Pakistan in hard or digital form. All individuals, firms, organizations or departments shall get their maps vetted and cleared from the Survey of Pakistan before use, publication, dissemination or circulation.]

17. Classified data.—The use and maintenance of all data or maps categorized as top secret, secret and restricted shall be in accordance with rules.

PART V

USE OF SURVEY MARKERS

18. Protection, use and maintenance of survey marks.—(1) Nobody shall damage, destroy or remove any survey mark or seize or occupy the land used for permanent surveying marks. In case the marks are established on private property, compensation thereof shall be paid by the ¹[concerned Division] under applicable law.

¹ Subs. and added by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) ss.7 and 8.

(2) No quarrying, demolition, soil gathering or removing or mining activities shall be carried out on, near or over any survey mark, which may damage or endanger its safety and effective utilization.

(3) No individual or organization shall carry out any type of construction within a radius of ten metres of a survey mark, except with prior permission of Survey of Pakistan and such individual or organization shall bear all expenses incurred on removal or re-establishment of the survey mark.

(4) Every surveyor or user of survey marks shall ensure its proper safety during the course of taking observations.

(5) The local administration shall be responsible for protection of all survey marks under its area of jurisdiction and shall send to Survey of Pakistan annual report on prescribed form about condition of the survey marks.

PART VI

NATIONAL AND INTERNATIONAL INITIATIVES

19. Geographical names.—(1) Notwithstanding anything contained in any other law for the time being in force, the ¹[concerned Division] shall establish, by notification in the official Gazette, a National Naming Authority which shall consist of not more than ten members and shall be chaired by the Surveyor General of Pakistan.

(2) The functions of the National Naming Authority shall be to appropriately name particular geographical locations and it shall have such powers to enable it to carry out the said functions as may be prescribed.

PART VII

OFFENCES AND PENALTIES

20. Offences.—Any organization, firm, individual or group of individuals engaged in any unauthorized activity within the meanings of this Act shall render itself an accused of illegal practices and

- (a) shall be asked by Survey of Pakistan to immediately suspend all such activities;
- (b) Survey of Pakistan shall ask concerned police to register a criminal case on such activities;

- (c) depending upon nature of such violations, the accused shall be asked by Survey of Pakistan or its designated official to deposit with the department of Survey of Pakistan the entire received money and a penalty up to fifty thousand rupees; and

¹Substituted by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.9.

- (d) in case Survey of Pakistan is convinced of major offence on part of the accused then his case shall be referred to a court of law and shall be tried for the following, namely:—
- (i) any individual or group causing hindrance and obstruction to the work being done by Survey of Pakistan and a registered surveyor shall be liable to be imprisoned for a term which may extend upto one month and a fine upto fifty thousand rupees;
 - (ii) any individual or group causing damage to survey mark shall be liable to be imprisoned for a term which may extend upto three months and a fine upto one hundred thousand rupees;
 - (iii) an individual who engages in geospatial data production, analysis and surveying and mapping activities in violation of the provisions of this Act shall be liable to be imprisoned for a term which may extend up to one year and a fine up to one million rupees;
 - (iv) every act of damaging, destroying, removing, seizing, occupying or establishing a mark resembling a permanent survey mark of Survey of Pakistan shall be an offence under this Act, punishable with imprisonment for a term which may extend up to one month and a fine up to fifty thousand rupees; ¹[*]
 - (v) the owners, directors or managers whosoever of any accused organization, firm, individual or group of individuals, which engages in geospatial data production, analysis, surveying and mapping activities or any related activities in violation of the provisions of this Act, shall, if it is proved that such activities have been committed with the consent or connivance on the part of the owner, director or manager, as the case be, be liable to be imprisoned for a term which may extend up to one year and a fine upto five million rupees ¹[; and]
 - ¹[(vi) any individual, firm, organization or department involved in printing, displaying, disseminating, using or circulating incorrect and unofficial version of map of Pakistan or any part of Pakistan in hard or digital form shall be liable to be imprisoned for a term which may extend up to five years or a fine of five million rupees or both.]

21. Jurisdiction of courts.—(1) Notwithstanding anything contained in the Code of Criminal Procedure, 1898 (Act V of 1898), no court other than a court of Magistrate of the First Class shall have jurisdiction to try an offence under this Act.

(2) The offences under this Act shall be non-cognizable and bailable.

¹ Omitted, subs. and added by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.10.

PART VIII

MISCELLANEOUS

22. Printing and publishing of maps and geospatial data.—(1) Any of the maps and geospatial data, whole or part thereof, falling in the responsibility of Survey of Pakistan, shall not be printed and published by any firm, individual or organization.

(2) During an emergency, selected printing presses of Pakistan shall come under command of Survey of Pakistan on required basis. A list of selected printing presses shall be prepared and updated, on annual basis, by Survey of Pakistan and notified by the Government of Pakistan.

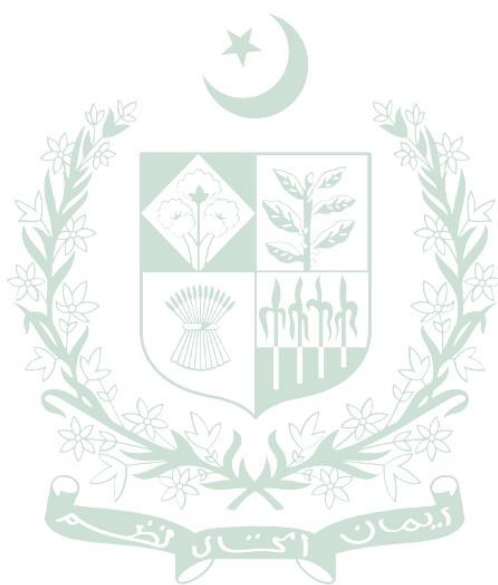
23. Power to make rules.—The ¹[concerned Division] may, by notification in the official Gazette, make rules for carrying out the purposes of this Act.

24. Indemnity.—No suit, prosecution or other legal proceedings shall lie against any person in respect of anything which is in good faith done or intended to be done under this Act.

25. Employees to be Civil Servants.—The terms and conditions of service under Survey of Pakistan shall be governed by the Civil Servants Act, 1973 (LXXI of 1973) and the rules made thereunder.

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¹Substituted by Surveying and Mapping (Amendment) Act, 2020 (Act No. XXXI of 2020) s.11.



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